What the MSA writes into an AF3-style trunk's pair representation, and why it helps mutation-effect prediction

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Abstract

Multiple sequence alignments (MSAs) have been progressively de-emphasized in protein structure prediction: AlphaFold2 made the MSA a first-class, recurrently processed input, whereas AlphaFold3 (AF3) and its open implementation RosettaFold-3 (RF3) compress MSA processing into a small module whose entire output passes once into the pair representation. Neither the AF3 nor the RF3 report measures what this channel contributes, and all existing MSA-perturbation evidence is at the input-to-output level. Here we measure, at the representation level, what the MSA writes into the pair representation of an RF3 trunk, and quantify its net effect on deep mutational scanning (DMS) prediction. Across three proteins, the MSA mildly compresses the per-mutant pair-representation ensemble toward a family consensus (effective rank −16% to −42%; linear CKA to the no-MSA representation 0.53–0.92), yet this compression *helps* cross-position mutation-effect prediction: the de-biased gain is +0.14–0.15 Spearman ρ on the two proteins where de-biasing was performed. A dose×content decomposition attributes the functional gain to per-column conservation (∼2/3) and pairwise covariation (∼1/3), saturating at ∼8 MSA rows, while a generic "shared-context tax" compresses rank without buying function. Gating the MSA to a single recycle installs ∼75% of the geometric and ∼85% of the functional effect, and nine subsequent MSA-free recycles wash out only ∼25%: the write is one-shot and near-irreversible, a causal confirmation of AF3's "all information passes through the pair representation" design. Roughly 60% of the MSA-specific functional advantage is reconstructible from the no-MSA pair representation plus ESM-2 embeddings. Across proteins, the functional gain tracks the magnitude of the representational rewrite, not raw MSA depth: family coherence, not depth, is the operative variable.

1 Introduction

The multiple sequence alignment entered protein structure prediction as the carrier of evolutionary covariation: residue pairs that co-vary across a family are likely to be in contact, and direct-coupling methods could fold proteins from MSAs alone, without any supervised structural data [8, 20]. AlphaFold1 converted coevolutionary statistics from an alignment into deep-learned distance potentials [7]; AlphaFold2 (AF2) made the MSA a first-class input, processed recurrently by the Evoformer alongside the pair representation, and trained with a masked-MSA reconstruction loss [1]. AF2's own analysis documented a dose–response with a characteristic threshold: accuracy drops substantially below ∼30 alignment rows and saturates beyond ∼100, and the authors hypothesized that MSA information is needed early, "to coarsely find the correct structure," rather than for refinement [1].

Since then, the field has steadily de-emphasized the MSA. ESMFold removed it outright, showing that a large protein language model (pLM) can replace MSA processing for most targets while being up to 60× faster [3]; ESM3 extended the single-sequence paradigm and its benchmarks show the residual MSA advantage concentrated on hard targets [4]. RFdiffusion dispenses with the MSA entirely, carrying the structural prior in pretrained weights instead [6]. AlphaFold3 kept the MSA but architecturally demoted it: MSA processing is reduced to four blocks with inexpensive pair-weighted averaging, and "the MSA representation is not retained and all information passes through the pair representation" [2]. RosettaFold-3 (RF3), an open AF3-style implementation, inherits this design [5]. Yet RF3's report contains no ablation of the MSA input—its only ablation statement concerns the MSA *search tool* [5]—and AF3's architectural demotion was likewise never accompanied by a measurement of what the MSA channel contributes inside the trunk.

In parallel, a substantial literature has characterized the MSA's influence at the input→output level. Perturbing the MSA—by depth subsampling, clustering, in-silico mutagenesis, or column masking—releases alternative conformations from AF2 [9, 11, 10, 12], and removing MSA features removes learned conformational-state biases in GPCR modeling [14, 15]. Conversely, MSA statistics remain a strong substrate for variant-effect prediction across depths [17], and MSA-based models retain variant-effect performance that single-sequence pLMs lose as they scale [19]. What is missing is a representation-level account: no study has quantified what the MSA writes *into the pair representation itself*—in terms of rank, representational similarity, or preservation of mutation-conditioned directions—nor decomposed the write into its information channels, nor tested causally whether the write is reversible, nor measured its net functional effect on mutation-effect prediction with depth-matched controls inside one fixed trunk.

We perform these measurements in RF3, using its pair representation (Z_{II}, captured at the distogram head) as the readout, on three proteins with DMS data spanning shallow to deep MSAs (15 to 512 rows). We ask four questions. (i) Geometrically, what does the MSA do to the ensemble of per-mutant pair representations? (ii) Functionally, does the write help or hurt cross-position mutation-effect prediction, and by how much after correcting for selection and preprocessing biases? (iii) What is the write made of—shared context, per-column conservation, or pairwise covariation—and when in the trunk's recycles is it installed? (iv) How much of the written content is already available from the no-MSA pair representation and from ESM-2 embeddings? We find that the MSA performs a one-shot, near-irreversible family-consensus write; that this write is functionally *beneficial* for DMS prediction; that conservation carries most of the benefit; and that the benefit scales with the magnitude of the rewrite rather than with alignment depth. We also report two methodological artifacts we encountered—a feature-slicing bug that initially inverted our headline conclusion, and a standardization-convention inconsistency—because both are silent, severe, and likely to affect similar analyses.

2 Results

2.1 The MSA mildly compresses the pair representation, and this helps prediction

We extracted the pair representation of RF3 for every single-point mutant of three proteins, with and without the MSA, slicing each mutant's representation at the true mutation site (row and column; Methods). The three proteins span three MSA regimes: the SARS-CoV-2 spike receptor-binding domain (RBD; 15-row shallow MSA, N=3998 mutants), human caspase-3 (CASP3; 69 Swiss-Prot homolog rows, homodimer), and the λ repressor CI domain (CI; 512-row deep UniRef30 MSA).

On all three proteins, the MSA compresses the per-mutant representation ensemble toward a family consensus, but mildly (Fig. 1, Table 1): the effective rank at 90% variance drops by 16–42% (RBD 231→171; CASP3 333→193; CI 61→51), the with/without-MSA representations remain

Table 1: The MSA pins the pair representation mildly and improves cross-position mutation-effect prediction on three proteins. eff90 = effective rank at 90% variance of the centered per-mutant feature ensemble; CKA = linear CKA between with-MSA and no-MSA representations; dCos = median cosine of per-mutant delta directions (MSA vs. no-MSA); $\Delta\rho$ = split-paired supervised gain ($\mathrm{sp}_{\mathrm{msa}} - \mathrm{sp}_{\mathrm{no}}$), best-epoch validation under overall standardization (optimistic; see §2.2); honest = de-biased estimate (§2.2). Splits = position splits with positive gain out of five.

Protein	MSA rows	eff90 no→MSA	CKA	dCos	$\Delta\rho$ (best-val)	Honest $\Delta\rho$
RBD	15	231 → 171 (−26%)	0.527	0.474	+0.179/+0.204 (5/5, 2 runs)	+0.14–0.15
CASP3	69	333 → 193 (−42%)	0.553	0.362	+0.162 (5/5)	+0.14–0.18
CI	512	61 → 51 (−16%)	0.917	0.844	+0.075 (4/5)	—

substantially aligned (linear CKA 0.527/0.553/0.917), and per-mutant delta directions (mutant minus wild type) are partially preserved (median cosine 0.474/0.362/0.844) while delta norms are amplified rather than suppressed (RBD ratio 1.21). This is a partial subspace compression with substantial representational continuity—not an erasure of mutation-conditioned structure.

Supervised probing against DMS labels shows that this compression is functionally beneficial (Fig. 3). Under the canonical split-paired protocol (Methods), the MSA-side representation outperforms the no-MSA side on all three proteins: RBD +0.179 and +0.204 Spearman ρ in two independent full-set runs (5/5 position splits positive in each; the 0.025 run-to-run spread matches our measured cross-process noise floor), CASP3 +0.162 (5/5), and CI +0.075 (4/5). The naive expectation that compressing per-mutant distinctions should destroy mutation information is wrong: the family-consensus write *re-encodes* mutation signals toward family evolutionary statistics, which is precisely the signal that must generalize across positions.

2.2 An honest effect size: de-biasing selection and preprocessing

The gains above are best-epoch validation scores under a standardization that pools all samples before splitting, and are optimistically biased by two independent routes. First, selecting the best epoch on a single validation set overfits it: in a dual-validation design (60/20/20 cross-position split), the selection optimism is +0.04 to +0.09 depending on the cell, worst for the MSA-only probe (+0.094, 5/5 splits). Second, fitting per-position standardization statistics on all samples leaks validation/test statistics into the transform; re-fitting them on training rows only ("strict") lowers the MSA-only probe from 0.644 to 0.589 on RBD (−0.055, 15/15 paired runs) while the no-MSA probe drops only 0.022.

Two independent de-biasing corrections converge on the same number. On RBD, held-out-test evaluation gives +0.141 (test at the val-selected epoch) and +0.143 (best test); strict train-only standardization gives +0.146. On CASP3, the dual-validation replication gives honest gains of +0.142 to +0.183 depending on estimator. We therefore report the **honest RBD MSA effect as +0.14–0.15 Spearman ρ** (honest absolute probe level $\rho \approx 0.59$), with the CASP3 effect in the same magnitude class (+0.14–0.18); the best-validation figures of §2.1 should be read as optimistic upper bounds. The RBD val-set asymmetry that drives part of the selection optimism is protein-specific: a pre-registered replication on CASP3 falsified the hypothesis that stronger geometric pinning causes it (CASP3 pins harder, −42% vs. −26%, yet shows no such offset), and which single stream is stable versus noisy inverts between the two proteins. Throughout, we treat effects within ±0.02 as uninterpretable—this is the measured cross-process noise floor of the pipeline (Methods).

Table 2: E1 dose×content arms (RBD, 500-mutant subset protocol). eff90 = effective rank at 90% variance; ρ = supervised cross-position Spearman (500-subset protocol, not comparable to the canonical full-set protocol); gain = ρ minus the depth01 baseline. The no-MSA input is *not* the honest baseline (see text).

Arm	Content	eff90	ρ	gain vs depth01
noMSA	none (sequence only)	181	0.414	−0.076
depth01	single row (baseline)	165	0.490	—
random	shared context, no biology	149	0.479	−0.011
rowshuffle	shared context, no biology	155	0.477	−0.013
consensus	profile only	156	0.501	+0.011
colshuffle	+ per-column frequencies	150	0.596	+0.106
depth04	4 real rows	157	0.536	+0.046
depth08	8 real rows	138	0.626	+0.136
full15	all 15 rows (with covariation)	139	0.638	+0.148

2.3 Content decomposition: a shared-context tax, conservation, and covariation

What in the MSA produces these effects? We constructed eight RBD MSA arms that orthogonalize dose and content: four real-alignment depths (1, 4, 8, and all 15 rows), a consensus arm (every row replaced by the consensus sequence, destroying variation but keeping the profile), a column-shuffle arm (per-column amino-acid frequencies preserved, covariation destroyed), a row-shuffle arm, and a random-sequence arm (Fig. 2, Table 2). Supervised numbers in this subsection come from the E1 dose-curve protocol (500-mutant subset, scalar standardization; Methods); arm comparisons are within-protocol.

Three channels separate cleanly. First, a **shared-context tax**: any alignment-like input—including random sequences—compresses the representation (effective rank 181 for no-MSA down to ∼149–155 for random/row-shuffled contexts). Presenting any shared per-residue context diverts representational capacity, but this buys *no* function: the honest zero-content baseline is the 1-row alignment (depth01, $\rho = 0.490$), not the no-MSA input (0.414)—the +0.076 gap between them is a featurization-path effect of presenting an alignment at all—and random or row-shuffled contexts sit at or below depth01 (0.479, 0.477). Second, **per-column conservation** carries about two-thirds of the biological gain: column-shuffled profiles (frequencies without covariation) reach +0.106 over depth01, of which the profile itself (consensus) contributes almost nothing (+0.011); the conservation-specific step from consensus to column-shuffled frequencies is +0.095. Consistent with this, the consensus and column-shuffle arms are nearly indistinguishable in rank (156 vs. 150) but separate sharply in how they preserve mutation directions (dCos 0.815 vs. 0.548). Third, **pairwise covariation** contributes the remaining third: restoring true rows (full15) adds +0.042 over column-shuffle. The dose response saturates early—depth04 +0.046, depth08 +0.136, full15 +0.148 over depth01, with effective rank plateauing from ∼8 rows—so the functional gain is monotone in conservation/covariation *content*, not in sequence count.

2.4 The write happens once and is near-irreversible

In RF3, as in AF3, the MSA enters the pair representation through outer-product-mean modules in the first four MSA blocks of each recycle; the pair representation is the only carrier thereafter. We tested the temporal structure of this write by gating MSA exposure to the first K of the network's ten recycles (Table 3). A single MSA recycle (K=1) installs ∼75% of the total effective-rank drop (181→150 of the 181→139 full effect) and ∼85% of the total functional gain ($\rho = 0.605$ of the way

Table 3: Write-once dynamics (E7, RBD). MSA exposure gated to the first K recycles. Geometric share = fraction of the full eff90 drop (181→139); functional share = fraction of the full ρ gain (0.414→0.638).

Arm	eff90	ρ (500-subset)	geometric share	functional share
noMSA	181	0.414	0%	0%
K=1	150	0.605	74%	85%
K=2	145	0.628	86%	96%
K=3	144	0.609	88%	87%
full15 (10 rc)	139	0.638	100%	100%

Table 4: Fusion triad, three proteins (best-validation, overall standardization). Single-stream probes (sp) and dead-stream-corrected marginals of FAESM over Z_{msa} and of Z_{no} over (Z_{msa}+FAESM). The CI FAESM marginal is driven by one outlier split (2/5 positive) and should be read as unresolved.

Protein	$\mathrm{sp}_{\mathrm{no}}$	$\mathrm{sp}_{\mathrm{msa}}$	$\mathrm{sp}_{\mathrm{faesm}}$	FAESM | Z_{msa}	Z_{no} | (Z_{msa}+FAESM)
RBD	0.465	0.644	0.484	−0.005	+0.002
CASP3	0.394	0.556	0.584	+0.051 (5/5)	−0.001
CI	0.374	0.449	0.452	−0.039 (noisy)	+0.008

from 0.414 to 0.638); K=2 and K=3 saturate the remainder. Nine subsequent MSA-free recycles wash out only ∼25% of the geometric effect: the trunk cannot self-correct the family-consensus write once installed. The near-equality of the K=1 and column-shuffle arms in effective rank (150 both) triangulates the mechanism: a single exposure already deposits the profile-plus-one-shot-covariation content, and repeated injection only deepens it marginally. This is a causal, representation-level confirmation of AF3's architectural statement that the MSA representation is not retained and all information passes through the pair representation [2]—and it sharpens AF2's dose–response observation [1]: not only is ∼8–15 rows enough, a single pass of processing is enough.

2.5 The MSA stream subsumes the no-MSA structural signal and duplicates ESM-2

We next asked how the MSA-written pair representation relates to the two other mutation-prediction signals available to the trunk setting: the no-MSA pair representation Z_{no} (structure without family context) and per-mutant ESM-2 650M embeddings (FAESM; evolution without structure). Using a validated fusion architecture with dead-stream controls (zeros streams; Methods), we computed each signal's marginal contribution over the others on all three proteins (Fig. 4, Table 4).

On RBD, Z_{msa} alone ($\rho = 0.644$) dominates: adding FAESM on top of Z_{msa} changes performance by −0.005, and adding Z_{no} on top of (Z_{msa}+FAESM) by +0.002—both inside the ±0.02 noise floor—while FAESM does carry +0.075 over a Z_{no} backbone. The MSA write thus subsumes the no-MSA structural signal and duplicates the ESM signal simultaneously on this protein. Cross-protein, the no-MSA marginal stays ≈ 0 everywhere (−0.001 CASP3, +0.008 CI), but the ESM duplication is protein-dependent: on CASP3, FAESM adds +0.051 over Z_{msa} (5/5 splits), so the MSA does not fully subsume ESM there; on CI the marginal (−0.039) is unresolved, driven by one outlier split at small N. Under strict train-only standardization, every fusion cell drops by 0.007–0.039 but all marginal conclusions are unchanged; only the absolute levels and the Z_{msa}-over-Z_{no} magnitude are convention-dependent.

The information accounting, however, is *opposite* to the optimization accounting. Late in training, the ESM encoder's gradient dominates on all three proteins (RBD gradient-norm ratios

$f/z \approx 6.5$, $f/m \approx 2.7$; CASP3 and CI $f/m \approx 3.5$–3.8), even where its information marginal is zero. We tested causally whether this gradient dominance suppresses the MSA stream's contribution on RBD with two interventions that rebalance gradients without adding information. Adding Gaussian noise to the ESM stream produces an inverted-U: validation ρ rises from 0.649 to a peak of 0.662 at 75% noise (paired +0.013) and returns to baseline at 100% noise (0.648, equal to the dead-stream control 0.655 within noise), while the late f/z gradient ratio collapses monotonically from 2.91 to 0.04. Slowing the ESM encoder's learning rate (content-cost-free) gives a plateau with the cleanest single arm at $\lambda = 0.25$ (+0.014, 5/5 splits). Both interventions recover the same small slice (+0.013–0.014) of the MSA stream's contribution. The reading is that on RBD the ESM stream is information-redundant with the MSA write yet optimization-harmful to it—gradient dominance is not information dominance.

2.6 The MSA write is partially reconstructible from no-MSA structure and ESM-2

If the MSA write is a family-consensus re-encoding, how much of it is already implicit in other features? We trained regressors from $(Z_{\text{no}}, \text{FAESM})$ to Z_{msa} across an architecture ladder (linear, MLP-256/512, 1–2-layer transformers; Fig. 5). Validation CKA against the real Z_{msa} rises from 0.74 (linear) to a plateau of $\sim$0.80–0.83 from MLP-512 (transformers do not exceed the MLP under best-epoch selection), with validation R^2 still climbing slowly (0.25→0.36). Functionally, a transformer-reconstructed Z_{msa} recovers $\sim$60% of the real MSA stream's prediction advantage over Z_{no} (ρ: Z_{no} 0.446, reconstructed 0.575, real 0.663), matching the geometric fidelity. The remaining $\sim$40% is MSA-specific content that is not a learnable function of the other two modalities at these scales.

Three controls sharpen the interpretation (Fig. 6). First, a depth ladder (2–8 transformer layers) with train–validation CKA gaps shows the high reconstruction fidelity is mostly generalization, not memorization (gap +0.09 at depth 8). Second, an orthogonal one-hot encoding of amino-acid identity—carrying only sequence, no evolution—reconstructs far worse at every depth (CKA $\sim$0.5, non-monotonic, with a train–validation gap that *grows* with depth), so the reconstructible content is genuine evolutionary information, not trivial sequence-to-structure memorization. Third, a naive concatenation architecture masks Z_{no}'s contribution (the 128-dim Z_{no} is swamped by the 1280-dim ESM stream); with separate per-modality encoders and CKA as the training objective, the hierarchy stabilizes across depth at both-streams 0.81 > ESM-only 0.75–0.76 > Z_{no}-only 0.72. The information hierarchy for reconstructing the MSA write is therefore: ESM (evolution) strongest, no-MSA pair structure a real but smaller contributor, pure sequence identity far behind—consistent with the ESMFold-era observation that language models absorb MSA-like information [3], now measured at the representation level.

2.7 The functional gain tracks the magnitude of the rewrite, not MSA depth

Across the three proteins, the supervised gain is ordered by how hard the MSA rewrites the representation, not by how deep the alignment is: RBD's shallow, coherent 15-row MSA produces the hardest consensus (lowest CKA 0.527) and the largest gain; CASP3's 69-row MSA compresses rank the most (−42%) with an intermediate gain; CI's deep-but-diverse 512-row MSA produces the softest consensus (CKA 0.917) and the smallest gain (+0.075). Within RBD, the arm series shows the same coupling: conservation-bearing arms rewrite mutation directions more (lower dCos) and predict better. Family *coherence*, not raw depth, is the operative variable—a refinement of the naive "deeper alignment = more information" assumption and a complement to AF2's $\sim$30-sequence threshold, which concerns family identification [1].

3 Discussion

3.1 Relation to prior work

Our results sit between two literatures that have largely talked past each other. The MSA-perturbation family showed, at the structure-output level, that the default MSA pins predictors toward single, family-consensus conformations—subsampling, clustering, or masking the MSA releases alternative states [9, 11, 10, 12], and removing MSA features removes training-set state biases [15, 14, 13]; fold-switcher studies find the default pipeline captures one of two conformers, sometimes via outright memorization that bypasses the coevolution channel [21, 22]. The variant-effect literature showed the opposite valence: MSA statistics are a consistently useful substrate across depths [17], and MSA-based models retain variant-effect performance that scaling single-sequence models sacrifices [19, 18]. Our representation-level measurements reconcile the two: the *same* family-consensus write that compresses the pair representation toward family statistics is the write that supplies the evolutionary signal DMS prediction needs. The compression and the benefit are not in tension; they are the same operation viewed geometrically and functionally.

For AF3-style trunks specifically, the write-once result (§2.4) converts an architectural statement—the MSA representation is not retained [2]—into a measured causal property: the write is installed in the first recycle and cannot be undone by the remaining recycles. The early saturation ($\sim$8 rows; §2.3) is shallower than AF2's $\sim$30-row threshold [1], though the comparison is between different objects (representation-level function in a frozen trunk vs. end-to-end structure accuracy). The partial reconstructibility of the write from ESM-2 embeddings (§2.6) gives a representation-level footing to ESMFold's inference that language models learn MSA-like information [3], and to ESM3's benchmark finding that the residual MSA advantage concentrates on hard targets [4]: here, even a 15-row MSA writes content that is only $\sim$60% recoverable from a 650M language model plus the no-MSA structure (mirroring at the representation level the output-level observation that the covariation channel is often dispensable even for complexes [16]). Practically, for DMS prediction with AF3-style features, an MSA—even a shallow one—is worth its cost; and where a pLM stream is fused with the MSA-bearing stream, its optimization dominance should be managed (noise or learning-rate pacing), since information redundancy does not imply optimization neutrality.

3.2 Methodological cautions: two silent artifacts

Two artifacts we encountered deserve emphasis because they are silent and likely general. First, *feature slicing*: an early version of our extraction saved the pair-representation row at position 0 for every mutant instead of the row at the mutation site. Because position 0 barely varies across mutants, the with-MSA ensemble looked artificially low-rank (effective rank 30 rather than 171) and the supervised signal looked destroyed (-40% rather than $+0.14$–0.15)—the contrast against the correctly sliced no-MSA features was entirely bogus. Any per-mutant structural feature pipeline should verify, explicitly, that the slice index tracks the mutation site; we now treat this as a mandatory sanity check. Second, *standardization conventions and evaluation hygiene*: three per-residue standardization conventions coexisted in our pipeline, and the scalar-per-position variant specifically degraded fusion cells ($+0.037$ spurious difference on one cell) while leaving single-modality probes robust; separately, fitting standardization statistics before splitting inflated the MSA-side probe by up to $+0.055$, and single-validation best-epoch selection added a further $+0.04$–0.09. None of these biases changed any qualitative conclusion—the triad ordering and all headline directions are convention-robust—but each is large enough to manufacture or destroy a publishable effect at the margins, and each was invisible until probed adversarially. We report de-biased numbers as primary (§2.2) and archive both conventions.

3.3 Limitations

Several boundaries qualify the claims. (i) A single trunk: all measurements are in RF3; although RF3 implements the AF3 architecture [5], trained-weight specifics may matter, and AF2-style trunks with a retained MSA track may behave differently. (ii) Three proteins, one label each, all single-point substitutions; the CI estimate rests on only 351 mutants and one of its marginals is unresolved. (iii) The de-biasing protocols were applied to RBD and CASP3; the CI gain (+0.075, 4/5) is a best-validation figure and may shrink similarly. (iv) The E1 decomposition was run on the RBD's shallow 15-row MSA; whether the conservation/covariation ratio holds for deep alignments is untested. (v) The functional readout is a small supervised head on frozen features; fine-tuning regimes could redistribute the benefit. (vi) The gradient-rebalancing gains (§2.5) are small (+0.013–0.014) and demonstrated on RBD only; the noise-interior arm is statistically soft (3/5 splits), with the learning-rate arm the cleaner evidence (5/5).

4 Methods

4.1 Proteins, DMS data, and MSAs

Three proteins with deep mutational scanning data were used. RBD: the SARS-CoV-2 spike receptor-binding domain (201-aa construct), ACE2-binding labels (bind_avg) from the yeast-display scan of Starr et al. [23], N=3998 single-point mutants. CASP3: human caspase-3 (homodimer, 244 positions per chain, 488 tokens), fluorescence activity labels from the microfluidic scan of Roychowdhury and Romero [24], 1469 single-point mutants. CI: the λ repressor CI domain (237-aa construct, 59 assayed positions), DMS_score labels from ProteinGym [17], N=351 mutants. MSAs: RBD uses a pre-existing curated alignment with 15 full-length rows; CASP3 uses 69 Swiss-Prot homolog rows and CI 512 UniRef30 rows, both built locally with MMseqs2.

4.2 Feature extraction

Pair representations were extracted from RF3 [5] by saving the pair tensor Z_{II} ($L \times L \times 128$) after the distogram head through an environment-gated hook in the model's forward pass, once per mutant, with and without the MSA. Per-mutant features are the row and column at the true mutation site, stacked ($2 \times L \times 128$); the slicing index was verified against known mutation positions after discovery of the position-0 slicing artifact (§3.2). ESM-2 650M (`esm2_t33_650M_UR50D`) final-layer per-residue embeddings [3] were computed per mutant (FAESM, $L \times 1280$).

4.3 Geometric metrics

Let $X \in \mathbb{R}^{N \times d}$ be the centered matrix of per-mutant flattened site features. Effective rank at 90% variance (eff90) is the number of principal components needed to explain 90% of total variance. Representational similarity between with-MSA and no-MSA ensembles is linear CKA [25]. Mutation-direction preservation (dCos) is the median cosine similarity, over mutants, between the with-MSA and no-MSA delta vectors (mutant minus wild type); norm suppression/amplification is the median ratio of delta norms.

4.4 Supervised probing

Two protocols are used and are not numerically interchangeable. The *canonical* protocol (three-protein geometry/function runs, fusion matrices, distillation): a small transformer head (SinglePredictorV2) on frozen features; per-position, per-channel z-scoring; cross-position 70/30 splits (5 seeds) with 3 head initializations (15 paired runs per cell); 300 epochs, AdamW (lr 10^{-4}, weight decay 0.05), cosine schedule, MSE loss, best-epoch validation Spearman ρ. Fusion cells use a cross-attention fusion block with an orthogonality penalty and dead-stream (zeros) controls on identical split$\times$init grids. The *E1/E7 dose-curve* protocol: a lighter per-position head, per-position scalar standardization, and a 500-mutant stride-8 subset of RBD; arm comparisons are made only within this protocol (control experiments place the convention effect on single-modality cells within ± 0.02). Recycle gating (E7) uses an environment variable that zeroes the MSA input after the first K recycles.

4.5 De-biasing and statistics

Strict standardization fits per-position, per-channel statistics on training rows only and freezes them onto validation/test. The *dual-validation* protocol uses 60/20/20 cross-position splits with the second validation fold as a held-out test. All reported effects are split-paired (paired across split$\times$initialization replicates). Effects within ± 0.02 Spearman ρ are treated as uninterpretable: this is the measured cross-process/cross-GPU noise floor of the pipeline. We use "consistent" only when the sign is shared by at least 3 of 5 (typically 5 of 5) paired splits.

4.6 Distillation (D1)

Regressors from $(Z_{\text{no}}, \text{FAESM})$ to Z_{msa} were trained under per-channel standardization (3 splits $\times$ 2 initializations, best-epoch validation): linear, MLP with 256/512 hidden units, and 1–8-layer transformers, in both naive-concatenation and separate-encoder (cross-attention) variants, the latter trained with a CKA objective; controls replaced one input modality or substituted a fixed orthogonal one-hot amino-acid codebook. Functional equivalence of reconstructed Z_{msa} was measured by training the standard head on reconstructed features.

Data availability

All analysis scripts, archived result files, and the claim-level provenance ledger accompanying this manuscript are available in the project repository (`msa_role_properties/`); embeddings will be deposited on request / upon publication.

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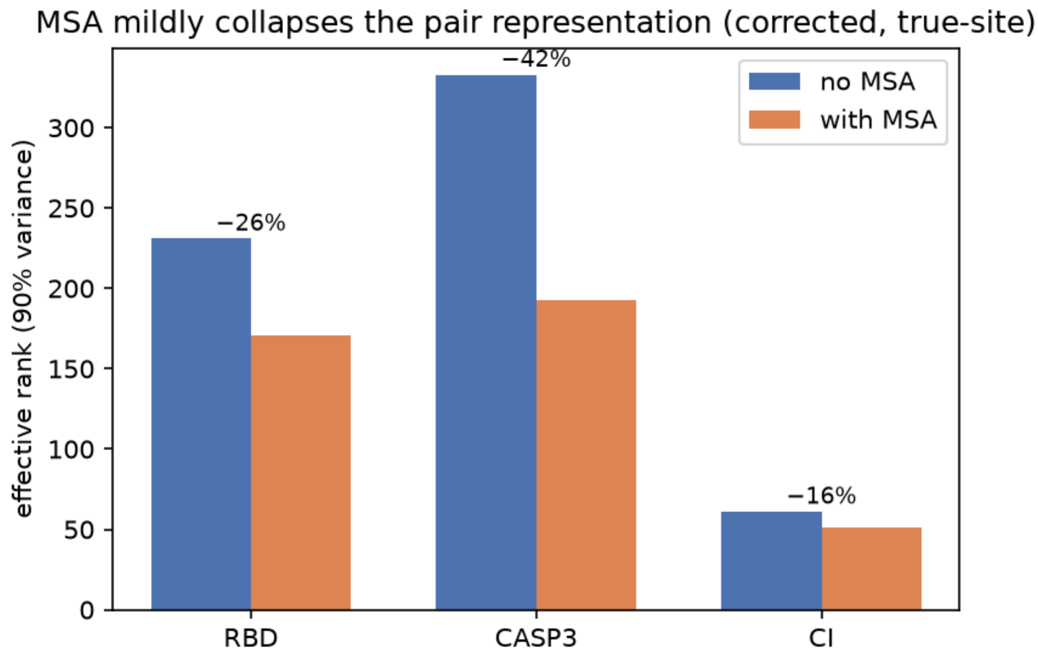

Figure 1: The MSA mildly compresses the pair representation on three proteins. Effective rank at 90% variance of the centered per-mutant site-feature ensemble, with and without the MSA (corrected true-site slicing, full mutant sets). Percentages give the relative rank reduction.

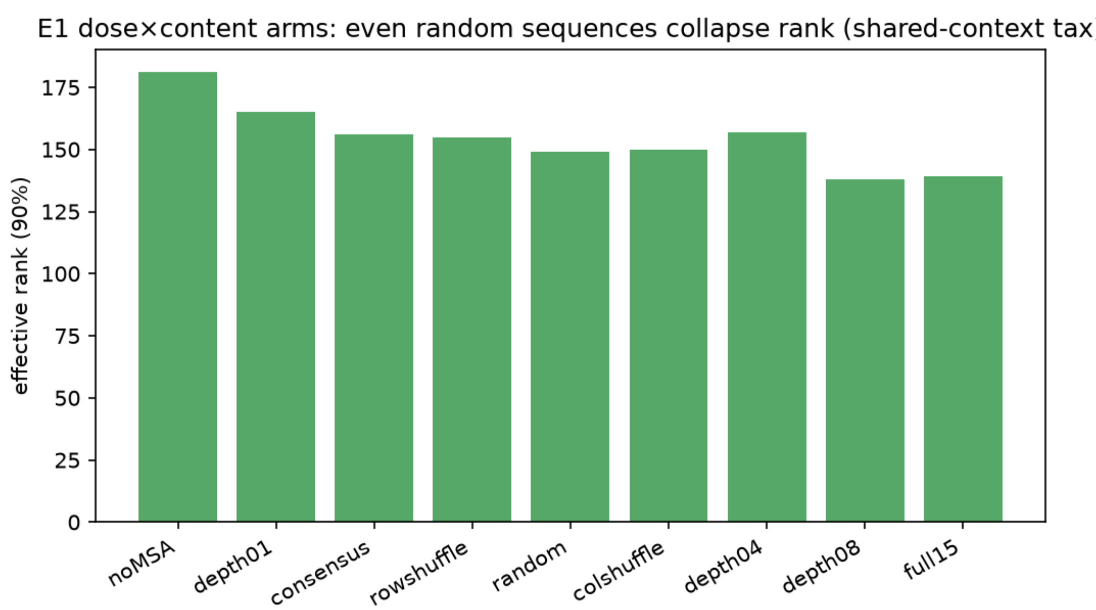

Figure 2: Effective rank across the E1 dose×content arms (RBD). Any alignment-like input—including random sequences—compresses the representation relative to no-MSA (the shared-context tax); the functional gain, by contrast, requires biological content (Table 2).

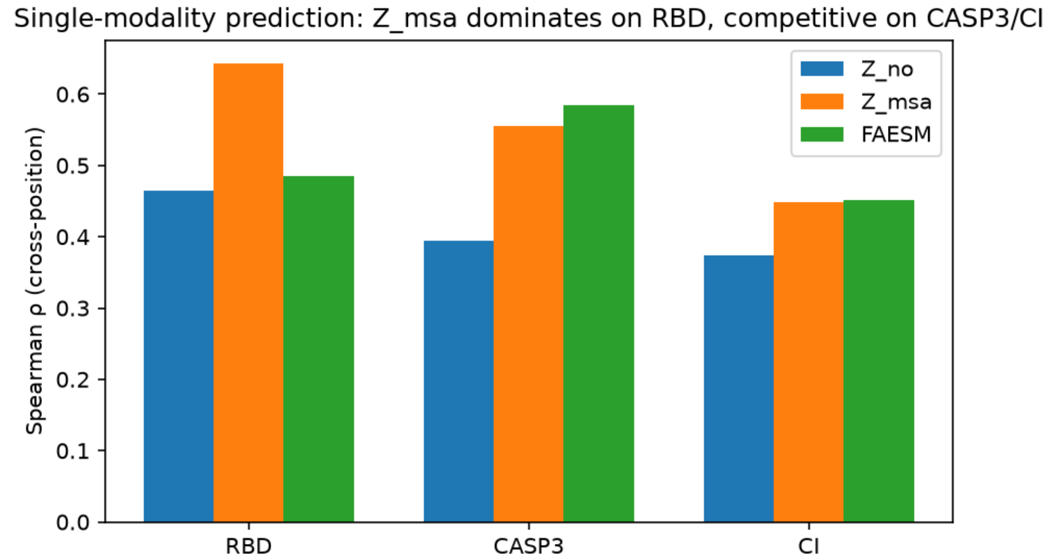

Figure 3: Single-modality cross-position prediction on three proteins. Best-validation Spearman ρ of probes on the no-MSA pair representation (Z_{no}), the with-MSA pair representation (Z_{msa}), and ESM-2 embeddings (FAESM). Values follow the optimistic best-validation convention; de-biased estimates are given in §2.2.

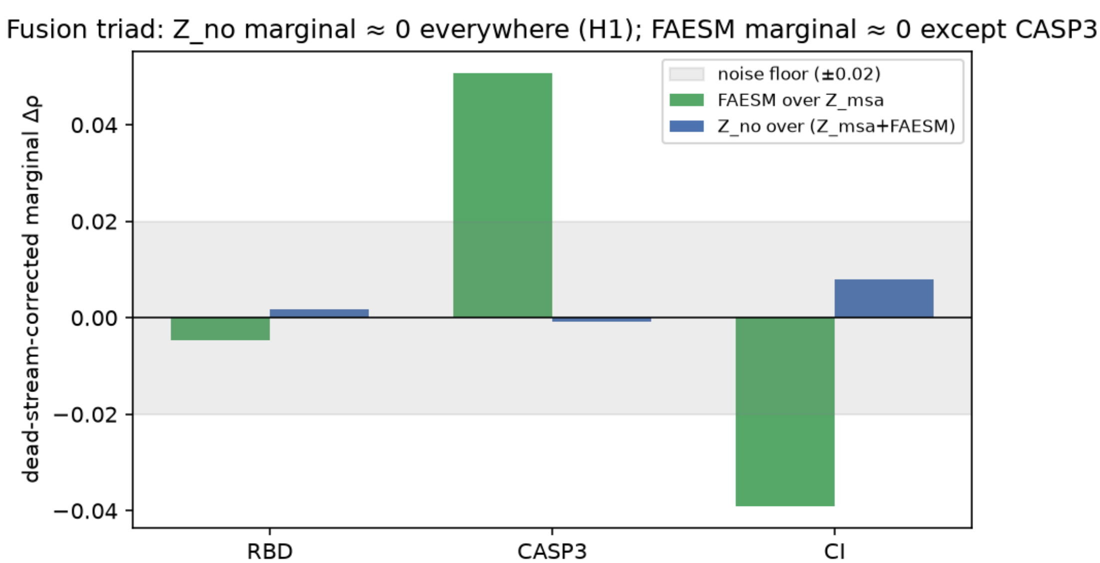

Figure 4: Dead-stream-corrected marginals of the fusion triad. The no-MSA stream's marginal over (Z_{msa}+FAESM) is ≈ 0 on all three proteins; FAESM's marginal over Z_{msa} is ≈ 0 on RBD, positive on CASP3, and unresolved (outlier-split-driven) on CI. The shaded band is the ± 0.02 noise floor.

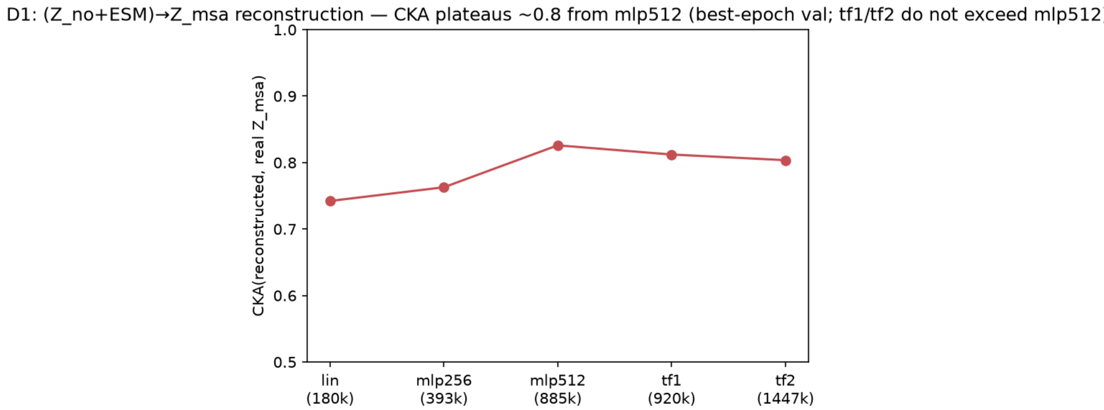

Figure 5: Reconstruction of Z_{msa} from $(Z_{\text{no}}, \text{FAESM})$ across an architecture ladder. Validation CKA plateaus at ∼0.80–0.83 from MLP-512 (transformers do not exceed it under best-epoch selection); parentheses give parameter counts.

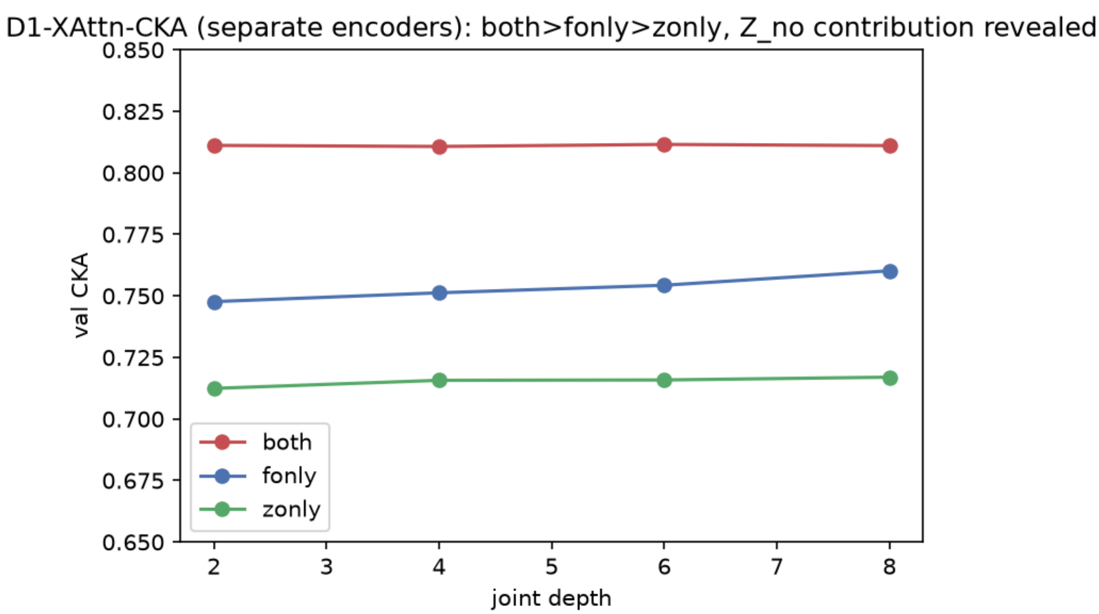

Figure 6: Separate-encoder distillation with a CKA objective. With per-modality encoders, the hierarchy is stable across depth: both streams > ESM-only > Z_{no}-only, showing that the no-MSA pair representation carries a real contribution that naive concatenation masks.